① Processos Estocásticos (Dia 28/10/2026)

Tema/Mote principal: $(dB_t)^2 = dt$.

No cálculo ordinário:

$$\Delta(x^2) = (x+\Delta x)^2 - x^2 = x^2 + 2x\Delta x + (\Delta x)^2 - x^2 = 2x\Delta x + (\Delta x)^2$$

$$\Rightarrow d(x^2) = 2x\,dx + \underbrace{(dx)^2}_{=0}$$

(em particular $\int_0^y d(x^2) = \int_0^y 2x\,dx$

$$y^2 = \int_0^y 2x\,dx$$

)

porém, no cálculo estocástico, $(dB_t)^2 = dt$. Então

$$\Delta(B_t^2) = (B_t + \Delta B_t)^2 - B_t^2 = 2B_t\,\Delta B_t + (\Delta B_t)^2$$

$$\Rightarrow d(B_t^2) = 2B_t\,dB_t + dt$$

$$\Rightarrow B_t^2 - B_0^2 = \int_0^t d(B_s^2) = \int_0^t 2B_s\,dB_s + \int_0^t ds$$

$$\Rightarrow B_t^2 = \int_0^t 2B_s\,dB_s + t$$

$$\Rightarrow \int_0^t B_s\,dB_s = \frac{1}{2}B_t^2 - \frac{t}{2}$$

(2)

$(dB_t)^2 = dt$ também "explica" a isometria de Itô:

$$\mathbb{E}\left(\left[\int_0^T \varphi_t dB_t\right]^2\right) = \mathbb{E}\left(\int_0^T \varphi_t dB_t \int_0^T \varphi_s dB_s\right)$$

$$= \mathbb{E}\left(\int_0^T\int_0^T \varphi_t \varphi_s \, dB_t dB_s\right) \quad (\text{FORMAL!})$$

$$= \int_0^T\int_0^T \mathbb{E}[\varphi_t \varphi_s dB_t dB_s] \; (*) \quad (\text{passo FORMAL!!})$$

~~incrementos independentes~~

FORMAL:

"$\forall s<t$, $\mathbb{E}[\varphi_t \varphi_s dB_t dB_s] = \mathbb{E}[\varphi_s dB_s \varphi_t \underbrace{\mathbb{E}[dB_t | \mathcal{F}_t]}_{=0}]$"

"$\mathbb{E}[dB_t|\mathcal{F}_t] = 0$ pois $dB_t \approx B_{t+\Delta t} - B_t$ com $\Delta t > 0$"

Logo, $(*) = \int_0^T\int_0^T \mathbb{E}[\varphi_t \varphi_s dB_t dB_s]$

$$= \iint_{[0,T]^2} \{s=t\} \, \mathbb{E}[\varphi_t \varphi_s dB_t dB_s] = \int_0^T \mathbb{E}[\varphi_t^2 (dB_t)^2]$$

$$= \int_0^T \mathbb{E}[\varphi_t^2 \, dt] = \int_0^T \mathbb{E}[\varphi_t^2] dt$$

$$\Rightarrow \mathbb{E}\left(\left(\int_0^T \varphi_t dB_t\right)^2\right) = \int_0^T \mathbb{E}(\varphi_t^2) dt$$

③ $(dB_t)^2 = dt$ também explica $V_2([0,T], B) = T$

$$V_2([0,T], B) \text{ "}=\text{"} \int_0^T (dB_t)^2 = \int_0^T dt = T.$$

Por quê estudar cálculo estocástico?

- Donsker (funcional): Dado $X_1, \dots, X_n$ iid, $\text{Unif}([0,1])$, Dado $g: \mathbb{R} \to \mathbb{R}$ com $\mathbb{E}(g(X_1)) = 0$. Então $\frac{1}{\sqrt{n}} \sum_{i=1}^{n} g(X_i) \xrightarrow{\mathcal{D}} N\left(0, \int_0^1 g^2 dx\right)$

reformulando: $\frac{1}{\sqrt{n}} \sum_{i=1}^{n} g(X_i) \xrightarrow{\mathcal{D}} \int_0^1 g(t)\, dB_t$

Essa formulação também vale conjuntamente pra $g \in G$, onde $G \subseteq L^2([0,1])$ não é muito grande.

- Donsker pra trajetórias: Dado $X_1, \dots, X_n$ iid, Seja $S_n = \sum_{i=1}^{n} X_i$

$$W^{(n)}(t) = \frac{1}{\sqrt{n}} \sum_{i=1}^{\lfloor nt \rfloor} X_i \quad \forall\, t \in [0,1]$$

$W^{(n)} \xrightarrow{\mathcal{D}} B$ (dentro do espaço de Skorokhod $D[0,1]$)

(4)

- Equações diferenciais Estocásticas:

EDO: $\frac{d}{dt} X_t = b(X_t)$

adicionamos uma fonte de ruído:

$$\frac{d}{dt} X_t = b(X_t) + \sigma(X_t) \frac{dB_t}{dt}$$

pra evitar falsa interpretação, escrevemos

$$dX_t = b(X_t)dt + \sigma(X_t) dB_t$$

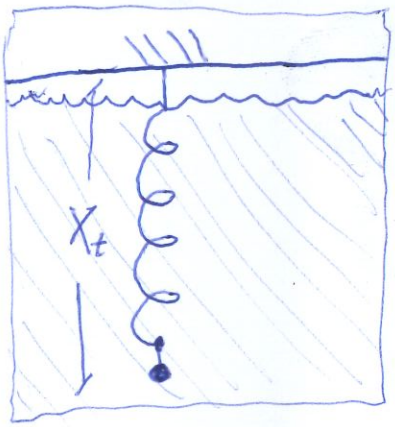
precisamente, isso significa que

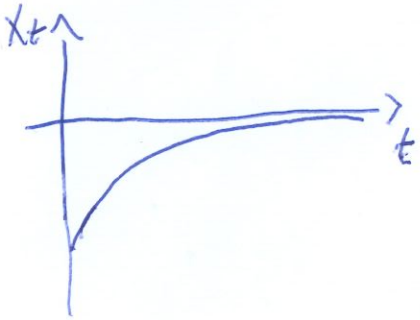
$$X_t - X_0 = \int_0^t b(X_s) ds + \int_0^t \sigma(X_s) dB_s$$

um exemplo concreto:

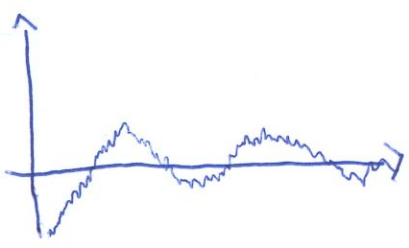
$$\frac{dX_t}{dt} = -\lambda X_t \quad , \lambda > 0$$

$$\Rightarrow X_t = X_0 \cdot e^{-\lambda t}$$





$$dX_t = -\lambda X_t dt + \sigma dB_t$$



(5)

Porquê usar B_t e não outro processo?

- Donsker / Universalidade
- Simplicidade dos processos Gaussianos
- Teorema (5.13 de Le Gall 2016) (Dambis-Dubins-Schwarz)

Seja M um martingal local contínuo, $q(t) = V_2([0,t], M)$ e suponha que $\lim_{t\to\infty} q(t) = \infty$. Então, existe um movimento Browniano $(B_t)_{t\geq 0}$ t.q. $M_t = B_{q(t)}$.

Última aula: Teo: $\forall \varphi \in \mathcal{H}^2$, $\exists X$ martingal contínuo t.q.

$$\mathbb{P}\left(X_t = I(\varphi \mathbb{1}_{[0,t]})\right) = 1 \quad \forall t \in [0,T].$$

Definição: $\int_0^t \varphi_s dB_s = X_t$.

prop: $\forall \varphi \in \mathcal{H}^2$, $0 \leq s \leq t$,

$$\mathbb{E}\left[\left(\int_s^t \varphi_u \, dB_u\right)^2 \,\middle|\, \mathcal{F}_s\right] = \mathbb{E}\left[\int_s^t \varphi_u^2 \, du \,\middle|\, \mathcal{F}_s\right]$$

⑥ Prova: Basta mostrar que $\forall A \in \mathcal{F}_s$

$$\mathbb{E}\left[\mathbb{1}_A \left(\int_s^t \varphi_u \, dB_u\right)^2\right] = \mathbb{E}\left[\mathbb{1}_A \int_s^t \varphi_u^2 \, du\right]$$

Tome $\tilde{\varphi} = \varphi \cdot \mathbb{1}_A \mathbb{1}_{(s,t]} \Rightarrow \tilde{\varphi} \in \mathcal{H}^2$

$\Rightarrow \|\tilde{\varphi}\|_{\mathcal{H}^2} = \|I(\tilde{\varphi})\|_{L^2}$ (*)

- $I(\tilde{\varphi}) = \int_0^T \varphi_u \mathbb{1}_A(\omega) \mathbb{1}_{(s,t]}(u) \, dB_u$

$= \mathbb{1}_A(\omega) \int_s^t \varphi_u \, dB_u$

$\Rightarrow \|I(\tilde{\varphi})\|_{L^2}^2 = \mathbb{E}\left[\mathbb{1}_A(\omega) \left(\int_s^t \varphi_u \, dB_u\right)^2\right]$ (**)

- $\|\tilde{\varphi}\|_{\mathcal{H}^2}^2 = \mathbb{E}\left[\int_0^T \varphi_u^2 \mathbb{1}_A(\omega)^2 \mathbb{1}_{(s,t]}(u) \, du\right]$

$= \mathbb{E}\left[\mathbb{1}_A(\omega) \int_s^t \varphi_u^2 \, du\right]$ (***)

(*) + (**) + (***) =

(7) Dados $\varphi, \psi \in \mathcal{H}$, τ tempo de parada, dizemos que

"$\varphi_t = \psi_t$ q.c. pra $t \leq \tau$" se $\exists \Omega_0 \subseteq \Omega$, $\mathbb{P}(\Omega_0) = 1$ t.q.

$\varphi_t(\omega) = \psi_t(\omega)$ $\forall (t,\omega) \in [0,T] \times \Omega_0$ com $t \leq \tau(\omega)$.

Prop: $\varphi, \psi \in \mathcal{H}^2$, τ tempo de parada. Assuma

$\varphi_t = \psi_t$ q.c. pra $t \leq \tau$.

Então $\int_0^t \varphi_s dB_s = \int_0^t \psi_s dB_s$ pra $t \leq \tau$.

Prova: Teorema 6.4 de (Michael Steele 2000) ∎

Localização

E se eu quisesse definir $\int_0^T \exp(B_t^4) dB_t$?

- $f(x) = \exp(x^4)$ é contínua
- mas $\mathbb{E}\left(\exp(B_t^4)^2\right) = \infty$.

Def: $\mathcal{L}^2_{loc}[0,T] := \left\{ \varphi \in \mathcal{M} : \mathbb{P}\left(\int_0^T \varphi_t^2 dt < \infty\right) = 1 \right\}$

($\mathcal{M}$: mensurável adaptado)

$\Rightarrow \mathcal{H}^2 \subseteq \mathcal{L}^2_{loc}$.

(8) Em particular, se $(\varphi_t)_{t\geq 0}$ é contínuo então $\varphi \in \mathcal{L}^2_{loc}$.

Def: $(\gamma_n)_{n\in\mathbb{N}}$ é uma sequência localizante pra φ se, $\forall n \in \mathbb{N}$,

(i) γ_n é tempo de parada

(ii) $\gamma_n \leq \gamma_{n+1}$

(iii) $\varphi_t^{(n)} = \varphi_t \mathbb{1}_{\{t \leq \gamma_n\}} \in \mathcal{H}^2$

(iv) $\mathbb{P}\left(\bigcup_{n\in\mathbb{N}} \{\gamma_n = T\}\right) = 1$

Teo: $\forall \varphi \in \mathcal{L}^2_{loc}[0,T]$, $\tau_n = \inf\left\{t \in [0,T] : \int_0^t \varphi_s^2 ds \geq n \text{ ou } t \geq T\right\}$

$\Rightarrow (\tau_n)_{n\in\mathbb{N}}$ é localizante pra φ.

Prova:

- $\tau_{n+1} \geq \tau_n$
- $\bigcup_{n\in\mathbb{N}} \{\tau_n = T\} = \left\{\int_0^T \varphi_t^2 dt < \infty\right\}$
- $\int_0^T (\varphi_t^{(n)})^2 dt = \int_0^T \varphi_t^2 \mathbb{1}_{\{\tau_n \geq t\}} dt \leq \int_0^{\tau_n} \varphi_t^2 dt \leq n$

~~precisa ser demonstrado usando o fato de τ_n ser tempo de parada~~

$\Rightarrow \|\varphi^{(n)}\|^2_{\mathcal{H}^2} \leq n$

$\Rightarrow (\tau_n)_{n\in\mathbb{N}}$ é localizante pra φ. ∎